Syed Tashfeenul Haque

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SUMMARY

ECE undergraduate with a solid foundation in C++ and Python, a strong inclination towards mathematics and analytical thinking. Actively studying LLM and transformer architectures — building models from scratch and applying concepts across real-world projects with a structured problem-solving mindset.

PROJECTS

Smart Solar Grass Cutter | Arduino ATmega328p, HC-SR04, IR Sensors, Bluetooth (HC-05), Proteus Sep 2025 – Oct 2025

- Designed and built an autonomous solar-powered grass cutter integrating a 20W solar panel with an ATmega328p microcontroller; achieved 6+ hours of uninterrupted operation per charge cycle with zero external power dependency.
- Implemented real-time obstacle detection using ultrasonic and IR sensors with a 30 cm detection threshold, cutting false-trigger rate to under 5% in field tests.
- Enabled wireless Android app control over Bluetooth with an effective range of ~10 m, reducing direct manual intervention by 80% compared to conventional mowers.
- Simulated and validated complete circuit in Proteus; recognized as the best project in the batch and highest-budget build at ₹6,000.

Agentic AI Loan Assistant | FastAPI, Streamlit, Google GenAI (Gemini), Python, asyncio EY Techathon

- Architected a 5-agent orchestration pipeline (Conversation -> KYC -> Credit -> Offer -> Document) with a FastAPI backend and Streamlit frontend themed for Tata Capital.
- Integrated Google GenAI (Gemini 2.5 Flash) via an asyncio.to_thread wrapper, achieving average agent response latency of ~2-3 s; reduced simulated loan intake cycle from ~25 min to under 9 min end-to-end.
- Implemented in-memory session persistence (optional Redis) and automated PDF sanction letter generation, cutting document turnaround from hours to <10 s.
- Built a real-time Agent Logs panel in the Streamlit UI with timestamps and color-coded badges for live orchestration monitoring across all 5 agents.

myChotuLLM | Python, PyTorch, Transformer Architecture *In Progress*

- *Building a decoder-only transformer from scratch in PyTorch — implementing scaled dot-product multi-head self-attention, learned positional encodings, and autoregressive text generation with greedy and top-k sampling.*
- *Experimenting with model scales from 1M to 15M parameters; targeting a character-level perplexity of <45 on a curated text corpus with ~10M tokens.*
- *Profiling training throughput and memory utilization to optimize batch size vs. sequence length trade-offs, achieving ~3x training speed-up with gradient accumulation on a single GPU.*

EDUCATION

Abdul Kalam Technical University (AKTU) / Galgotias College of Eng. & Tech. Aug 2024 – May 2028 (Expected)
B.Tech – Electronics and Communication Engineering
Relevant Coursework: Data Structures & Algorithms, Digital Electronics, Signals & Systems, Embedded Systems, Engineering Mathematics, Microcontrollers

SKILLS

- **Languages:** Python, C++, JavaScript, TypeScript, SQL, C
- **ML & AI:** PyTorch, Transformers, LLM Architecture, Google GenAI (Gemini), Multimodal AI
- **Backend & Frameworks:** Node.js, Express, FastAPI, REST APIs, WebSockets
- **Frontend:** React, HTML, CSS, Tailwind CSS, Vite
- **Databases:** MongoDB, MySQL, Drizzle ORM
- **Tools:** Git, GitHub, Postman, Proteus, VS Code, Linux